(ii) Short-cut method

The arithmetic mean can be calculated by using any arbitrary value A as origin and write d as the deviation of the variable X then,

$$\overline{X} = A + \frac{\Sigma d}{n} \text{ where } d = X - A.$$

b) Grouped data

(i) Direct method

The formula for computing the mean is $\overline{X} = \frac{\Sigma fx}{N}$

where f is frequency, X is the variable, $N = \Sigma f$ i.e. total frequency.

(ii) Short-cut method

The arithmetic mean is computed by applying the following formula:

$\overline{X} = A + \frac{\Sigma fd}{N}$, where A is assumed mean, $d = X - A$, $N = \Sigma f$

Arithmetic mean for Continuous case

The arithmetic mean may be computed by applying any of the following methods:

(i) Direct method

(ii) Short-cut method

(iii) Step deviation method

(i) Direct method

When direct method is used arithmetic mean is defined as

$$\overline{X} = \frac{\Sigma fm}{N}$$

Where m = midpoint of each of the class interval,

f = the frequency of each class interval

$N = \Sigma f$ = total frequency

(ii) Short-cut method

The arithmetic mean is computed by applying the following formula.

$$\overline{X} = A + \frac{\Sigma fd}{N}$$

where A is assumed mean (or) arbitrary value, $d = m - A$ is deviations of mid-point from assumed mean and $N = \Sigma f$

(iii) Step Deviation Method

In case of grouped (or) continuous frequency distribution, the arithmetic mean is

$\overline{X} = A + \left(\frac{\Sigma fd}{N} \times c\right)$, where $d = \frac{(m - A)}{c}$

A is any arbitrary value (or) assumed mean and c is the magnitude of class interval.



All the above three methods of finding arithmetic mean in continuous case gives us the same answer.

Mode:

Mode is the value which repeats maximum number of times among the given observations.

Median:

Median is exactly a middle value and it exceeds and exceeded by the same number of observations. Median is one of the positional measure. Some other related positional measures are also described below.

NOTE



It is believed that the students might be familiar with the above concepts and our present syllabus continues from the following.

8.1.2 Related Positional Measures - Quartiles, Deciles and Percentiles :

Besides median there are other measures which divide a series into equal parts. Important amongest these are quartiles, deciles and percentiles.

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